

TE Curve Verification Pipeline Familywise ONNX Report - wave3_3_raw_offset_curve_aware

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_offset_curve_aware/2026-07-12-09-53-26__track2_wave3_3_raw_offset_curve_aware_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_offset_curve_aware/2026-07-12-09-53-26__track2_wave3_3_raw_offset_curve_aware_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_offset_curve_aware/2026-07-12-09-53-26__track2_wave3_3_raw_offset_curve_aware_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave3_3_raw_offset_curve_aware/2026-07-12-09-53-26__track2_wave3_3_raw_offset_curve_aware_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave3_3_raw_offset_curve_aware` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave3_3_raw_offset_curve_aware_fw_simplified_setpoints</code>	<code>2026-07-11-22-26-25__te_wave3_3_raw_offset_curve_aware_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave3_3_raw_offset_curve_aware_bw_simplified_setpoints</code>	<code>2026-07-11-22-59-48__te_wave3_3_raw_offset_curve_aware_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave3_3_raw_offset_curve_aware_global_simplified_setpoints</code>	<code>2026-07-11-21-48-20__te_wave3_3_raw_offset_curve_aware_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/forward/2026-07-11-22-26-25__te_wave3_3_raw_offset_curve_aware_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/forward/2026-07-11-22-26-25__te_wave3_3_raw_offset_curve_aware_fw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=132-val_mae=0.00358121.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/backward/2026-07-11-22-59-48__te_wave3_3_raw_offset_curve_aware_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/backward/2026-07-11-22-59-48__te_wave3_3_raw_offset_curve_aware_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=145-val_mae=0.00357142.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/global/2026-07-11-21-48-20__te_wave3_3_raw_offset_curve_aware_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/global/2026-07-11-21-48-20__te_wave3_3_raw_offset_curve_aware_global_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=159-val_mae=0.00354446.ckpt</code>

Aggregate Metrics

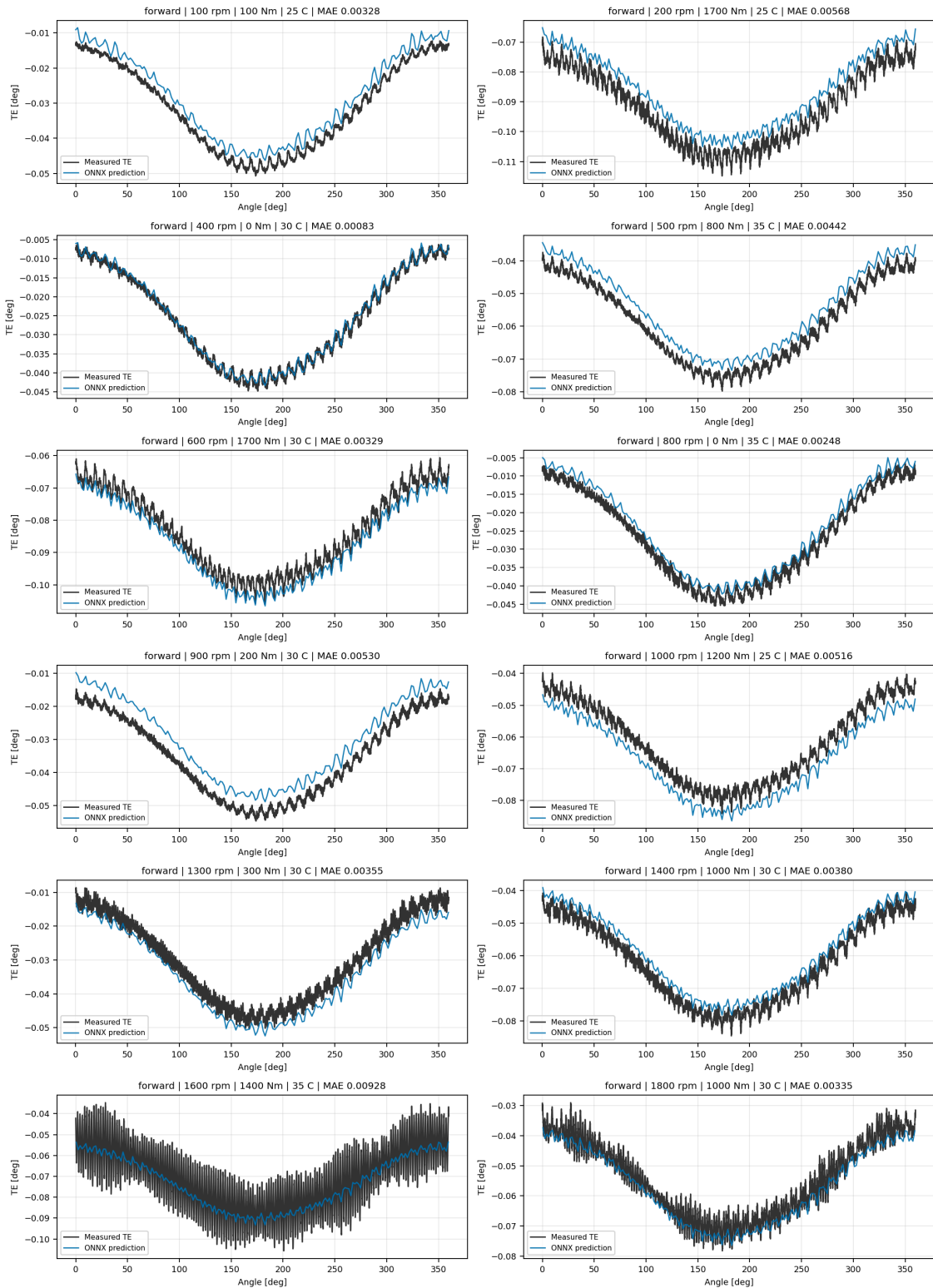
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003421	0.003699	8.001	13.729
backward	97	0.003497	0.003820	8.108	14.896
global	194	0.003501	0.003784	8.161	14.778

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000170	0.003056	0.001285	0.003107
backward	0.000460	0.002923	0.001538	0.003334
global	0.000111	0.002970	0.001388	0.003407

Forward 12-Curve Page

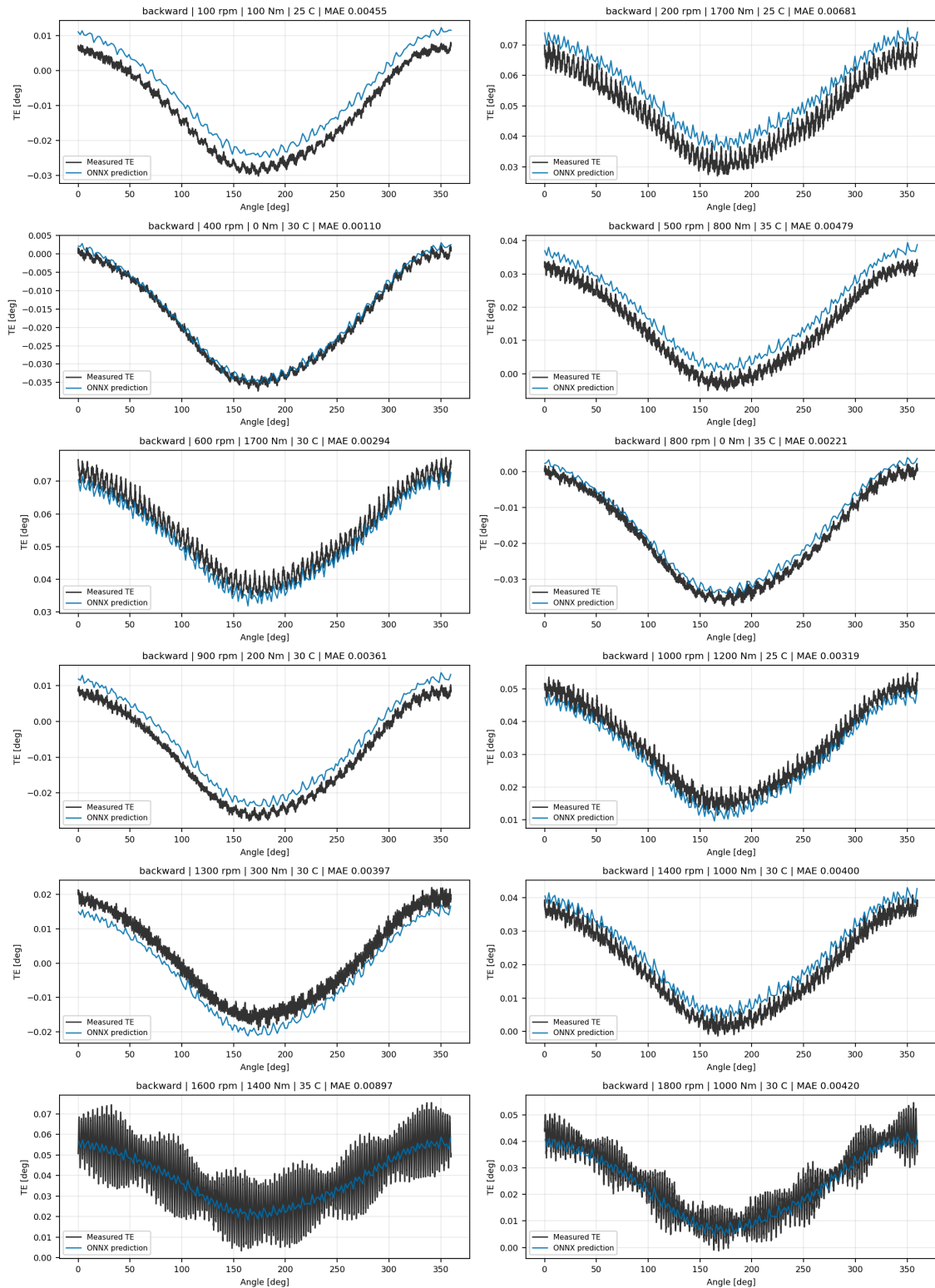
Simplified Dataset + Setpoints | wave3_3_raw_offset_curve_aware | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

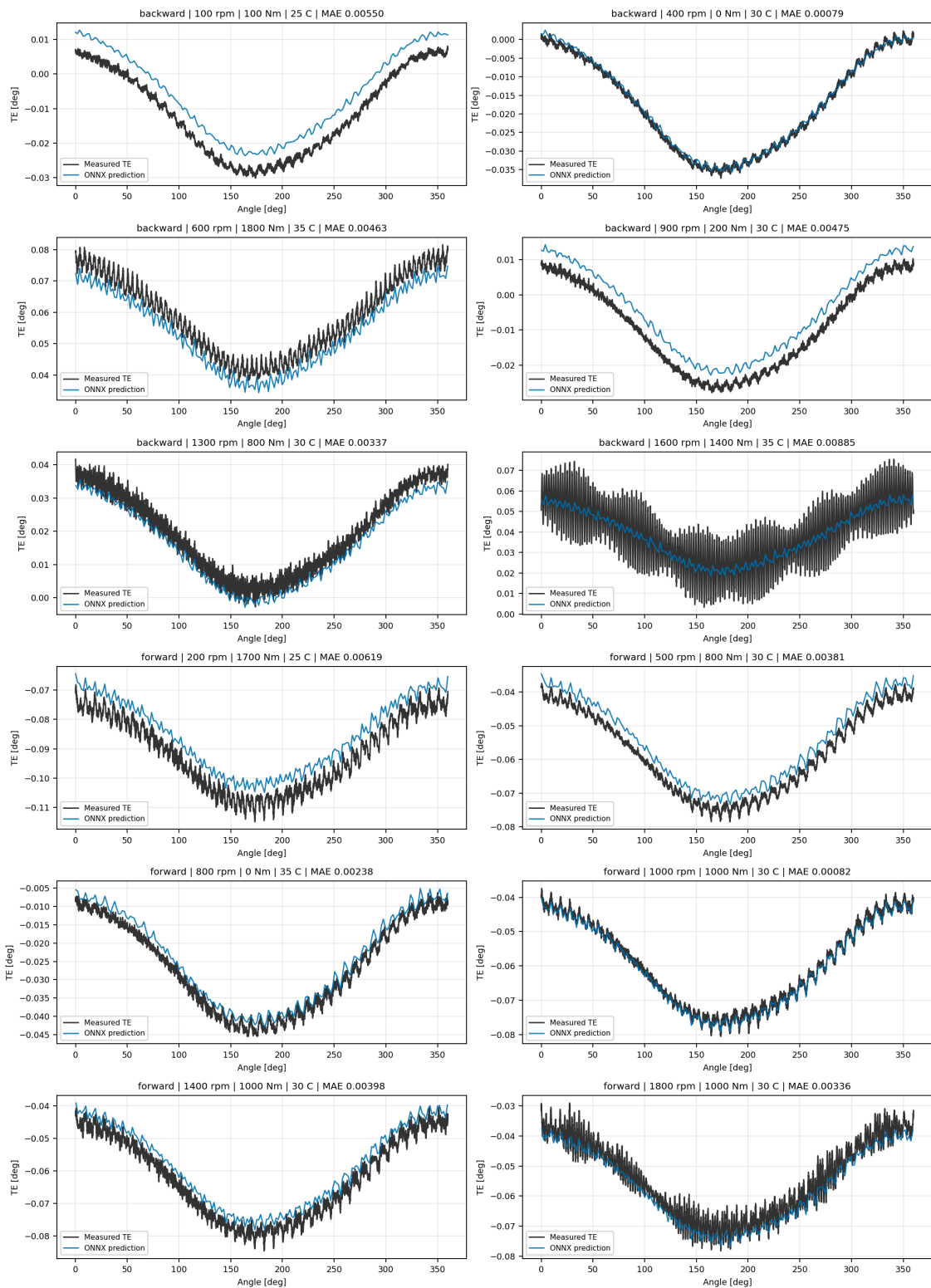
Simplified Dataset + Setpoints | wave3_3_raw_offset_curve_aware | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave3_3_raw_offset_curve_aware | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave3_3_raw_offset_curve_aware ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_3_raw_offset_curve_aware_fw_polished_setpoints	2026-07-12-00-23-59_te_wave3_3_raw_offset_curve_aware_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave3_3_raw_offset_curve_aware_bw_polished_setpoints	2026-07-12-00-59-31_te_wave3_3_raw_offset_curve_aware_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave3_3_raw_offset_curve_aware_global_polished_setpoints	2026-07-11-23-47-11_te_wave3_3_raw_offset_curve_aware_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/forward/2026-07-12-00-23-59_te_wave3_3_raw_offset_curve_aware_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/forward/2026-07-12-00-23-59_te_wave3_3_raw_offset_curve_aware_fw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=119-val_mae=0.00195047.ckpt
backward	models/polished_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/backward/2026-07-12-00-59-31_te_wave3_3_raw_offset_curve_aware_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/backward/2026-07-12-00-59-31_te_wave3_3_raw_offset_curve_aware_bw_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=112-val_mae=0.00201121.ckpt
global	models/polished_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/global/2026-07-11-23-47-11_te_wave3_3_raw_offset_curve_aware_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave3_3_raw_offset_curve_aware/global/2026-07-11-23-47-11_te_wave3_3_raw_offset_curve_aware_global_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=122-val_mae=0.00189514.ckpt

Aggregate Metrics

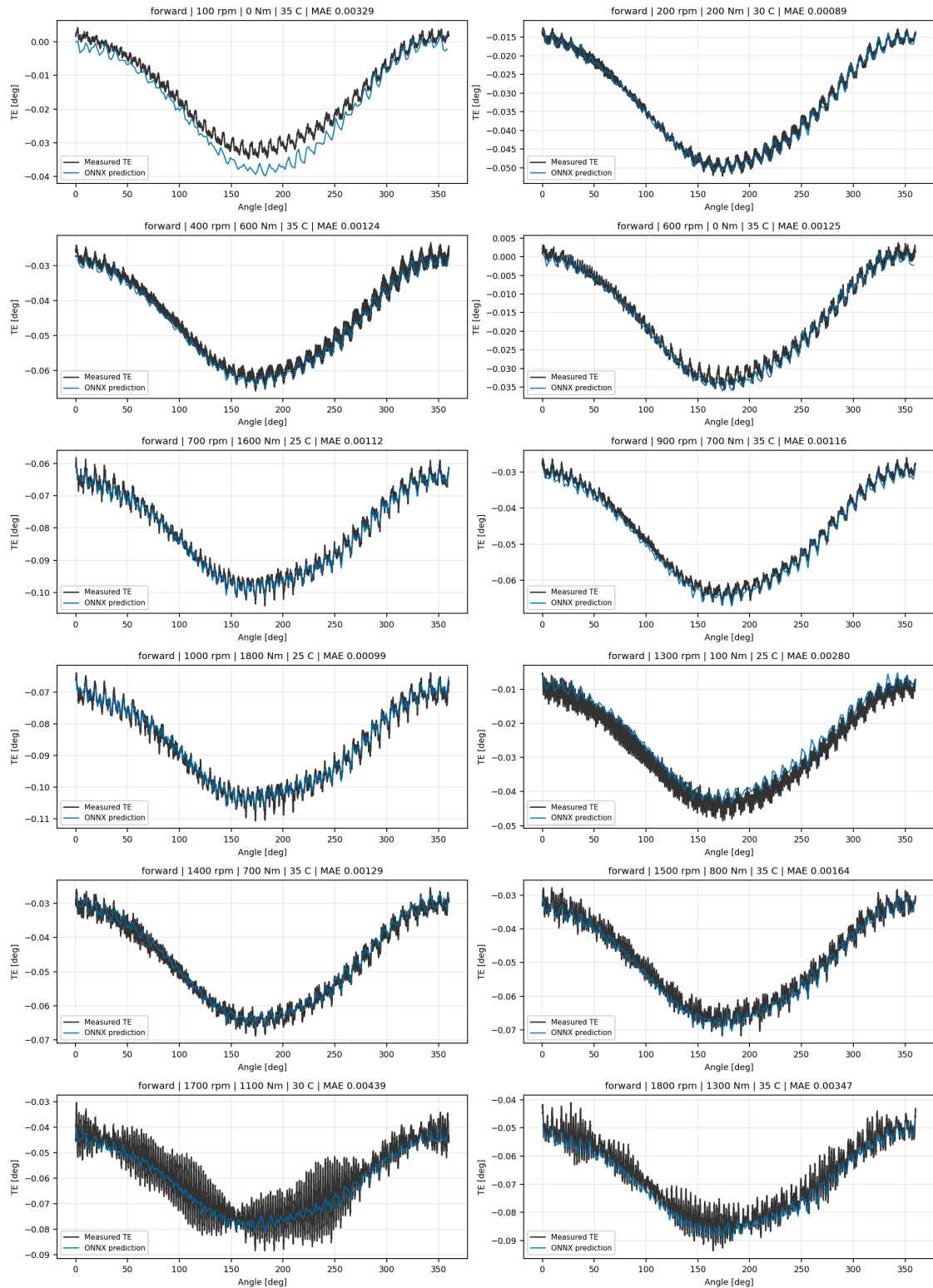
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.002048	0.002433	4.530	9.846
backward	94	0.002561	0.003002	4.711	10.783
global	194	0.002203	0.002608	4.389	10.783

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000515	0.001236	0.001578	0.003517
backward	0.000251	0.001103	0.002102	0.005874
global	0.000162	0.001002	0.001821	0.004606

Forward 12-Curve Page

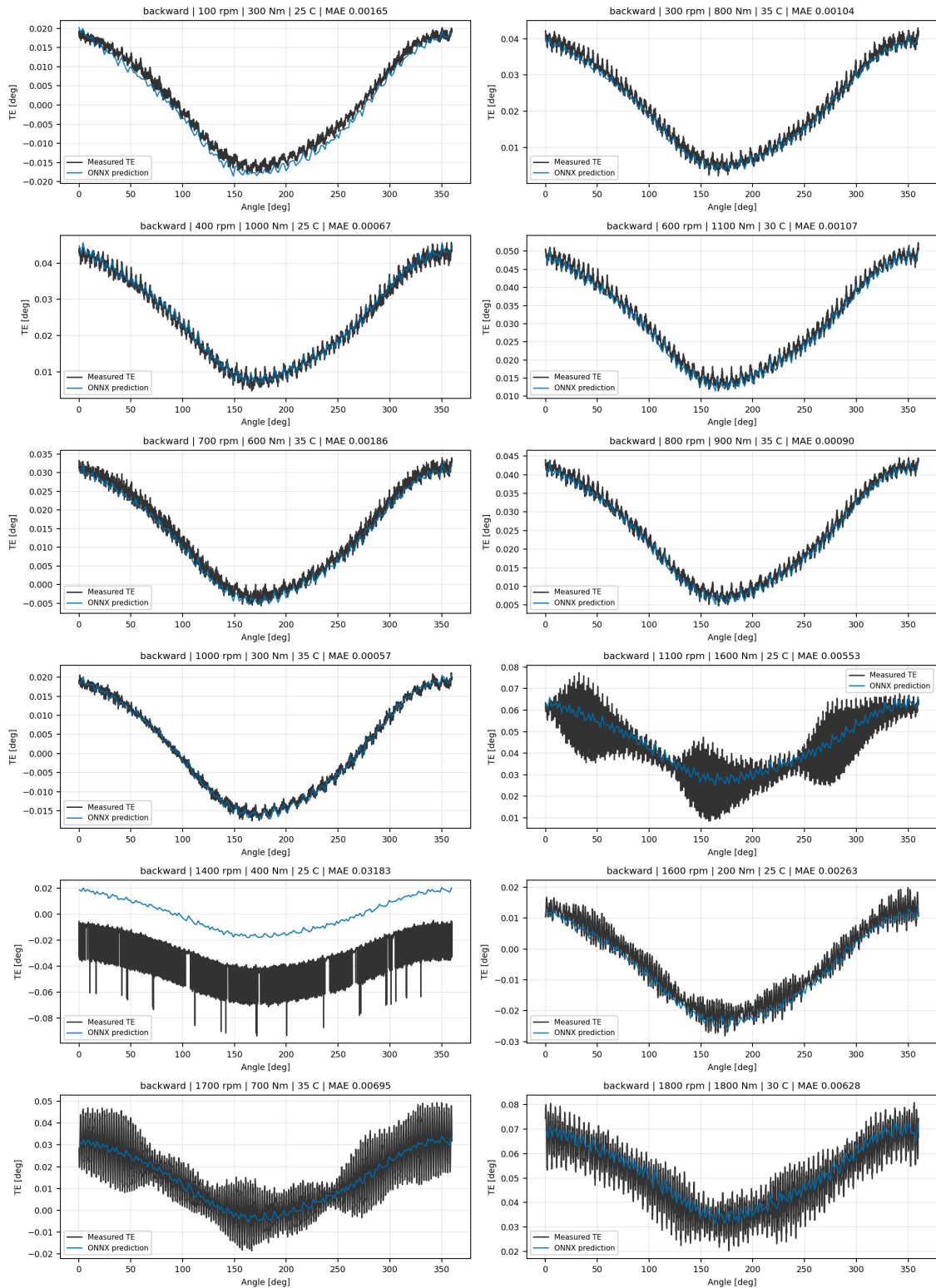
Polished Dataset + Setpoints | wave3_3_raw_offset_curve_aware | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

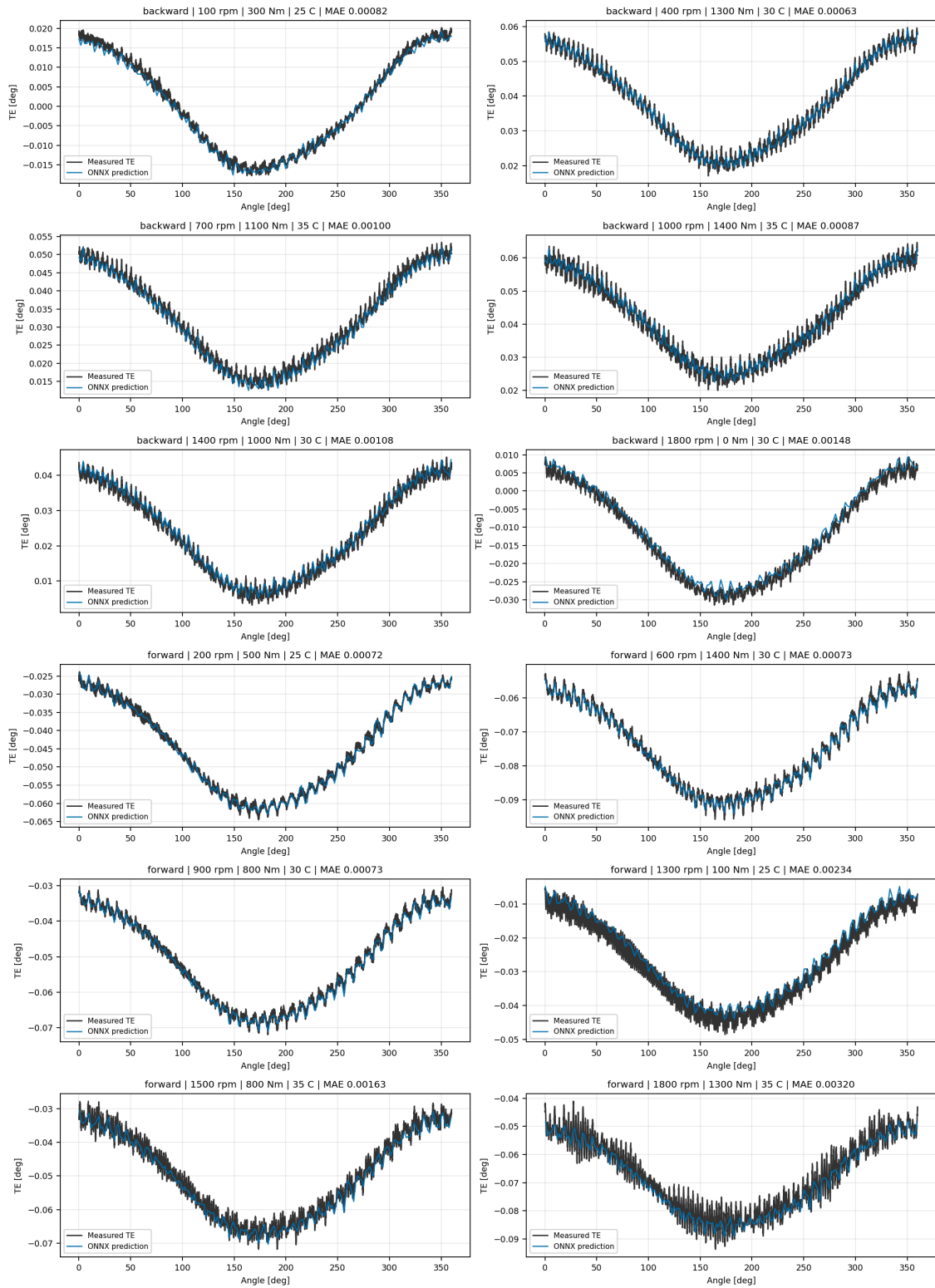
Polished Dataset + Setpoints | wave3_3_raw_offset_curve_aware | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave3_3_raw_offset_curve_aware | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave3_3_raw_offset_curve_aware ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave3_3_raw_offset_curve_aware_fw__polished_actual_values	2026-07-12-02-24-38__te_wave3_3_raw_offset_curve_aware_fw__polished_actual_values	polished_point_v1
backward	te_wave3_3_raw_offset_curve_aware_bw__polished_actual_values	2026-07-12-02-57-47__te_wave3_3_raw_offset_curve_aware_bw__polished_actual_values	polished_point_v1
global	te_wave3_3_raw_offset_curve_aware_global__polished_actual_values	2026-07-12-01-40-59__te_wave3_3_raw_offset_curve_aware_global__polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave3_3_raw_offset_curve_aware/forward/2026-07-12-02-24-38__te_wave3_3_raw_offset_curve_aware_fw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_3_raw_offset_curve_aware/forward/2026-07-12-02-24-38__te_wave3_3_raw_offset_curve_aware_fw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=103-val_mae=0.00192801.ckpt
backward	models/polished_dataset/actual_values/exported/wave3_3_raw_offset_curve_aware/backward/2026-07-12-02-57-47__te_wave3_3_raw_offset_curve_aware_bw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_3_raw_offset_curve_aware/backward/2026-07-12-02-57-47__te_wave3_3_raw_offset_curve_aware_bw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=199-val_mae=0.00184997.ckpt
global	models/polished_dataset/actual_values/exported/wave3_3_raw_offset_curve_aware/global/2026-07-12-01-40-59__te_wave3_3_raw_offset_curve_aware_global__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave3_3_raw_offset_curve_aware/global/2026-07-12-01-40-59__te_wave3_3_raw_offset_curve_aware_global__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=155-val_mae=0.00187021.ckpt

Aggregate Metrics

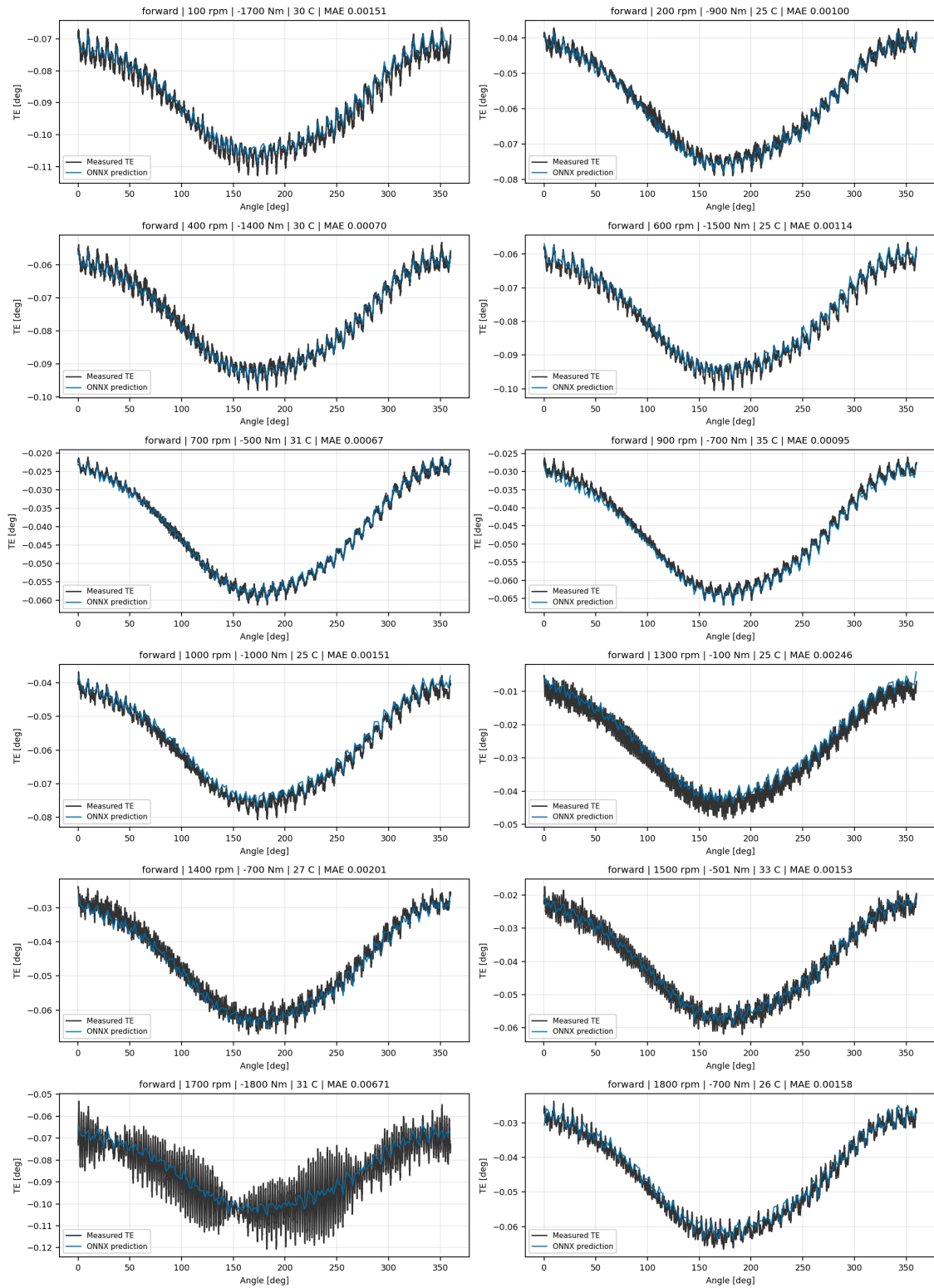
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001940	0.002313	4.284	9.624
backward	94	0.002175	0.002632	4.119	10.422
global	194	0.002001	0.002411	4.069	10.192

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000025	0.000969	0.001586	0.003856
backward	-0.000091	0.000516	0.002086	0.005351
global	-0.000033	0.000671	0.001817	0.004590

Forward 12-Curve Page

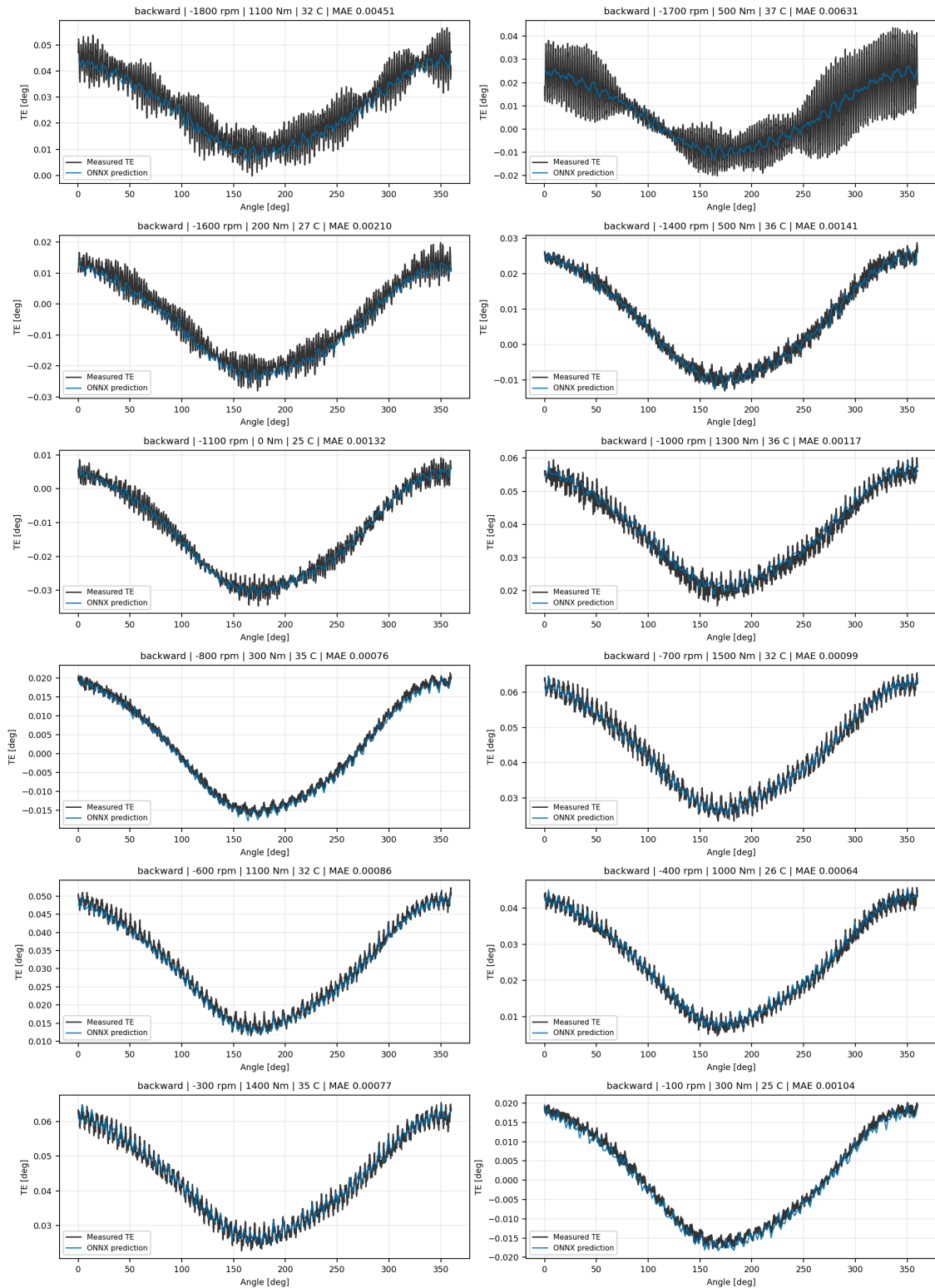
Polished Dataset + Actual Values | wave3_3_raw_offset_curve_aware | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

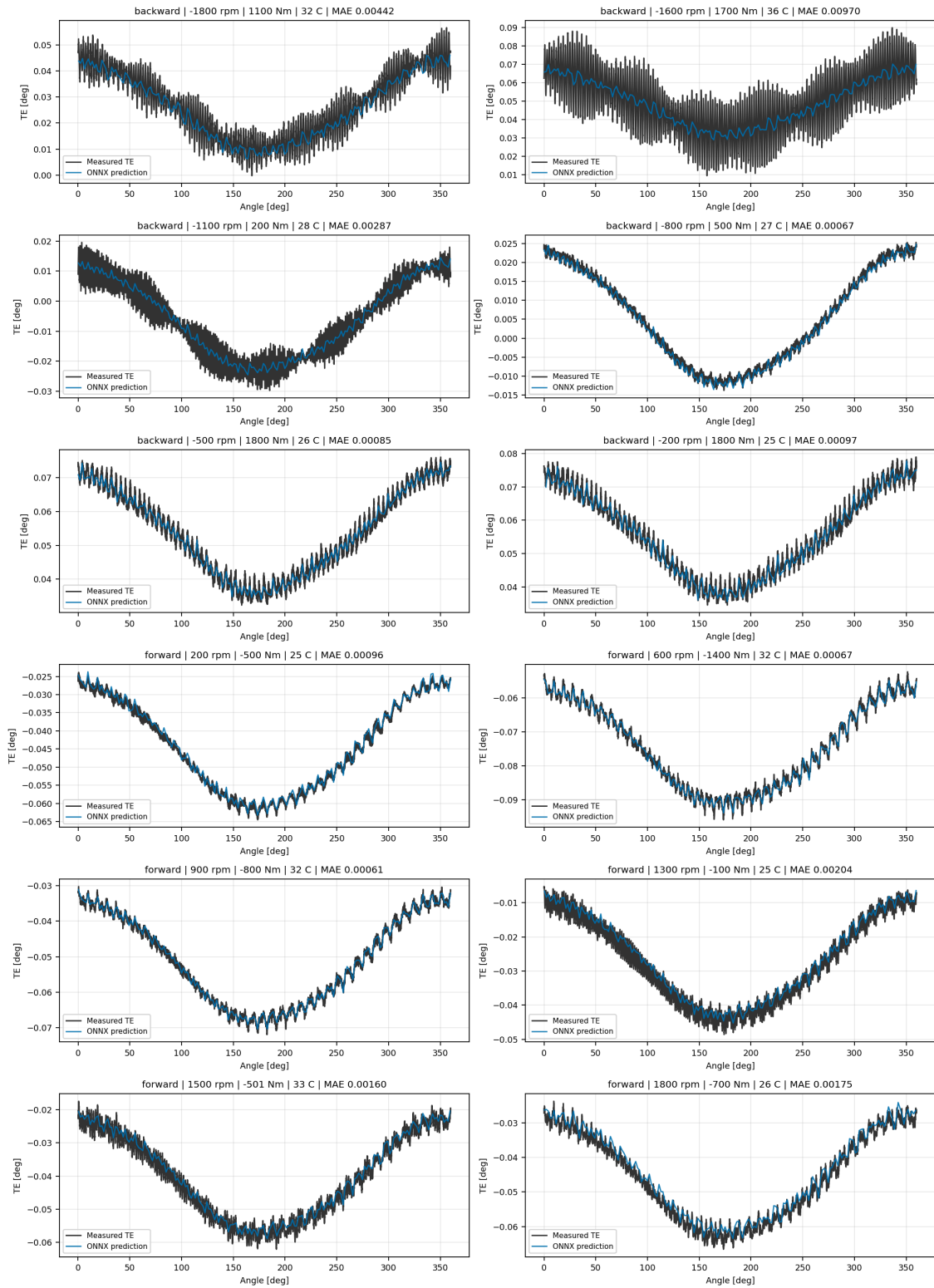
Polished Dataset + Actual Values | wave3_3_raw_offset_curve_aware | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave3_3_raw_offset_curve_aware | global



polished_dataset_actual_values global 12-curve collage